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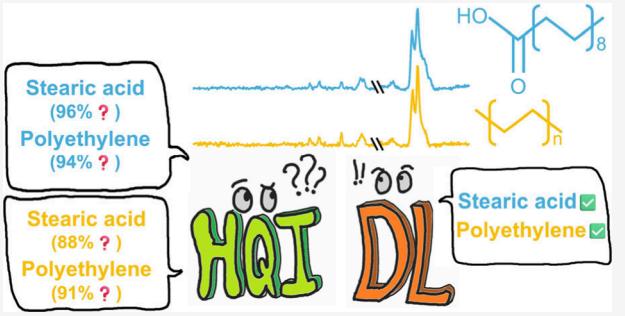
Reducing Spectral Confusion in Microplastic Analysis: A U-Net Deep Learning Approach

Jeonghyun Lim, Juhui Seo, and Dongha Shin*

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ABSTRACT: Among the various analytical techniques that have been proposed with the growing significance of microplastic detection, Raman spectroscopy is a powerful technique for detecting microplastics. However, the structural similarity in Raman spectra between fatty acids and polyethylene (PE) frequently causes misclassification by HQI-based methods, particularly when analyzing environmental samples containing mixed fatty acids. Herein, a U-net-based deep learning model was employed to precisely classify PE, stearic acid (SA), oleic acid (OA), mixtures of SA and OA, sodium dodecyl sulfate (SDS), and polypropylene based on their Raman spectra. Additionally, by incorporating a binarization technique commonly utilized in material chemistry, high scalability for both qualitative and quantitative analyses is provided. Consequently, the U-net model achieved accuracy improvements over the Pearson correlation coefficient of 2.05% to 11.09% for spectra with high signal-to-noise ratio (SNR) and 21.21% to 48.97% for spectra with nonaveraged spectra. Additionally, it demonstrated at least 36.69% higher accuracy compared to metrics such as Spearman correlation coefficient, cosine similarity, and Manhattan/Euclidean distance. This deep learning-based approach significantly reduces the confusion between PE and fatty acids observed in conventional Raman spectral analyses of microplastics, thereby demonstrating its potential applicability in microplastic standardization and analysis fields.



INTRODUCTION

As the growth in awareness of the risks posed by microplastics has resulted in a significant increase in related research,[1,2] leading to the numerous methodologies proposed for microplastic detection.[3] Mass-based techniques are useful for quantitatively analyzing microplastics, however, they are destructive and incapable of providing size analysis for individual microplastic particles. In the case of electron microscopy, while it allows for high-resolution imaging, it requires ultrahigh vacuum conditions, making in situ analysis challenging. FTIR is a nondestructive and rapid technique that is widely used for initial screening of microplastics; nevertheless, its inherent limitations make it difficult to accurately detect microplastics in water samples or those smaller than 10 μm.[4] Compared to these analytical techniques, Raman spectroscopy is attracting increasing attention in environmental fields because it is less affected by aqueous environments, allows for nondestructive measurement of smaller particles, and provides valuable molecular vibrational information.

Meanwhile, a common approach for identifying microplastics using Raman spectroscopy involves a matching procedure between acquired spectra with a reference database. For this purpose, HQI-based classification techniques have been generally employed, which involve comparing an acquired spectrum against spectra in a database and then identifying it as the most similar entry. HQI serves as the accuracy metric in this procedure; a material is confirmed as classified if the HQI for the most similar database class surpasses a predetermined threshold. Examples of these HQI-based techniques include: Pearson correlation coefficient[5] (PCC) method, Spearman correlation coefficient[6] (SCC) method, Cosine Similarity[7] (CS), Euclidean distance[8] (ED), and Manhattan distance[9] (MD).

However, the reliability of these hit quality index (HQI)-based methods is being called into question, particularly for the detection of polyethylene (PE). PE is expected to be one of the most prevalent types of microplastics among the most widely used plastics worldwide.[10,11] Specifically, PE consists of a repeating $-CH_2-$ chain structure, which closely resembles the structure of naturally occurring fatty acids. This structural similarity can lead to confusion in spectroscopic analyses and may result in inaccurate assessments of the environmental distribution of PE particles.

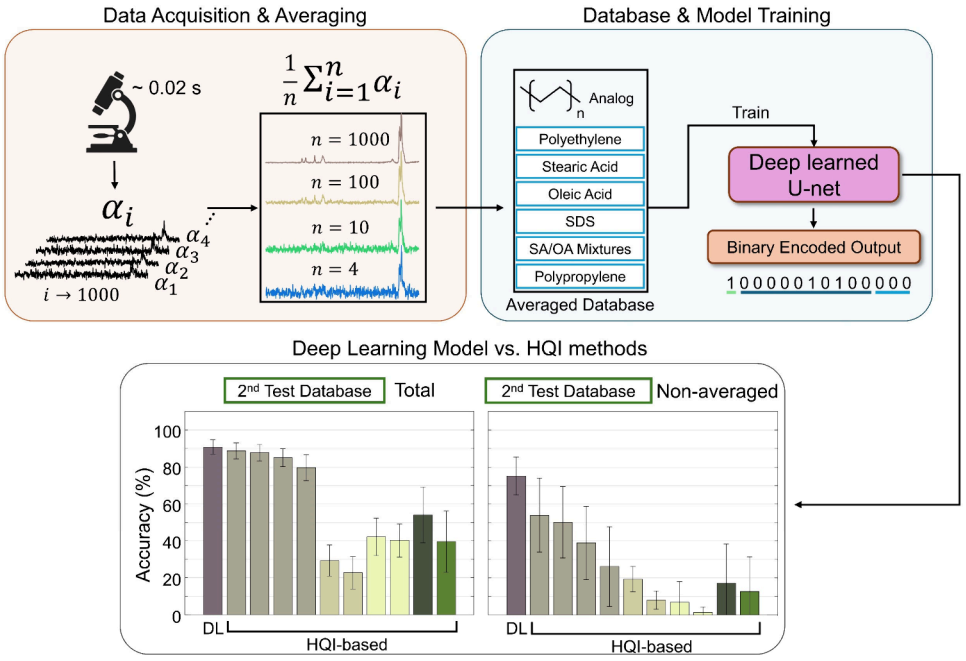
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Scheme 1. Workflow for Data Acquisition, Database Construction via Averaging, and Subsequent Model Training[a]



[a]Spectra for each CH_2-CH_2 chain analog material were acquired with a 0.02-second exposure time, yielding 1,000 spectra per material. Following this, a spectral database with progressively enhanced Signal-to-Noise Ratios (SNR) was assembled by grouping the nonaveraged spectra in sets of *n*. This database was then utilized to train a Deep Learning U-Net model, which produces a binary encoded output. The resulting model exhibits superior accuracy when compared to HQI-based methodologies.

The problem was discussed in an annex of the International Organization for Standardization document,[12] which discussed the standardization of microplastic measurement with vibrational spectroscopy. Additionally, this risk of misidentification has been researched. Witzig et al.[13] reported that stearates, fatty acids, and sodium dodecyl sulfate (SDS) present in disposable gloves could be misinterpreted as PE during analyses using Raman spectroscopy, FTIR, and pyrolysis−gas chromatography/mass spectrometry (pyr−GC/MS). Similarly, Li et al.[14] demonstrated the possibility of fatty acids being misclassified as PE in Raman spectra. Notably, their research emphasized the issue of HQI values for fatty acid Raman spectra exceeding the commonly accepted reliability thresholds, highlighting the fundamental limitations of the method.[15−17]

Several solutions have been proposed to address this misidentification issue, but they are insufficient. For example, adding fatty acids to the database or using alcohol pretreatment to remove them[14] were suggested as solutions, however, environmental samples often contain larger fatty acids alongside minor ones, and small amounts of alcohol are insufficient for complete removal. Thus, there is a need to develop analytical methods capable of more precisely distinguishing PE from various fatty acids.

Against this backdrop, advancements in AI technology offer promising solutions for the sophisticated classification of complex Raman spectra.[18,19] Subsequent studies demonstrated the potential for qualitative and quantitative Raman spectral analysis with high accuracy using deep learning models.[20−25] However, these studies predominantly employed one-hot encoding, which assigns each material to a separate position in the output. This approach has the drawback of requiring an increase in output size as the number of substances to classify grows. Additionally, prior research has been predicated on analyses of high-SNR spectra.[21−23] Nevertheless, given the exceedingly small size and abundance of microplastics, extended exposure times are impractical for their analysis.[26] Therefore, it is crucial to conduct qualitative analysis utilizing low-SNR spectra acquired under short exposure conditions.

Herein, we present a U-net-based deep learning strategy, enhanced by binarization and low-SNR spectrum augmentation, designed as a generalizable framework for precise microplastic classification in environmental mixtures. To begin, we selected a set of materials to test the limits of spectroscopic differentiation. PE was selected as the primary microplastic target. Stearic acid (SA) was selected as the primary spectral confounder. As a common, long-chain saturated fatty acid, the molecular structure of SA is dominated by the same $-CH_2-$ chain as PE, making it the weakness point for spectral overlap and the ultimate test of a classification model's specificity. To further assess the robustness and versatility of our approach, we expanded our data set to include oleic acid (OA), an unsaturated fatty acid commonly found in nature, polypropylene (PP), another prevalent polyolefin with a structure analogous to PE, and SDS, a frequent laboratory contaminant that can be misidentified as PE.

Further, we utilized the U-net architecture—a model known for its high performance in Raman spectral analysis[27−30]—and applied an output binarization technique common to materials chemistry.[31,32] With these techniques, the scalability of the output for qualitative and quantitative analysis is enhanced. Lastly, by averaging the model augmentation based on spectra with low SNR, we constructed a model capable of analyzing low SNR spectra.

As a result, the trained model demonstrated an accuracy improvement of 2.05% to 11.09% for spectra with high signal-to-noise ratio (SNR) and 21.21% to 48.97% for spectra with

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nonaveraged spectra, compared to PCC. Additionally, the model outperformed the SCC, CS, MD, and ED by at least 36.69% in accuracy. These results validate the model's ability to accurately distinguish between PE and fatty acids, offering a significant advancement over conventional HQI-based classification methods. Scheme 1 presents the process of database construction, the U-net model architecture, the binary-encoded output, and the corresponding analysis results.

EXPERIMENTAL SECTION

Bulk Materials. PE was obtained from Korea Petrochemical Ind. (South Korea). SDS was obtained from Tokyo Chemical Ind. (Japan). PE and SDS were supplied in powder state and used as received. The 15 mL conical centrifuge tube was used as PP obtained from Neurex Life Science (South Korea). SA was obtained from DUKSAN Pure Chemicals Co., Ltd. (South Korea). OA was obtained from DAEJUNG Chemicals & Metals Co., Ltd. (South Korea). All substances were acquired in their pure state.

The spectra of PE and SDS were acquired in the powder state, the spectra of PP were acquired in the bulk state, and the spectra of SA and OA were acquired in the capillary tube (DWK Life Sciences, United States).

To make the mixture of fatty acids, the fatty acids were weighed to make up a total of 0.01 mol in each mixture. The molecular weight of SA was calculated to be 284.484 g/mol, and the molecular weight of OA was calculated to be 282.468 g/mol. In addition, the density of OA was calculated to be 0.895 g/mL since it exists as a liquid at room temperature (25 °C). Each mixture was prepared with mole fraction changes of 10% as shown in Figure 2.

The measured fatty acids melted within the 15 mL conical centrifuge tube in the water bath. After melting, the mixture was stirred at 800 rpm with a magnetic bar for 5 min. The mixture was mixed with a vortex mixer for 30 s, additionally.

Simulated Environmental Sample with Polyethylene MP Inclusion. Bulk polyethylene is ground in its bulk state using a milling machine under cryogenic conditions to generate irregularly shaped debris (6775 Freezer/Mill Cryogenic Grinder, SPEX SamplePrep, U.S.). The irregularly fragmented PE was mixed with olive oil (composed of 70% oleic acid[33,34]) and ethanol to form a dispersion. This mixture was drop-cast onto a silicon wafer. The particle sizes were measured under bright-field (BF) microscopy, and Raman spectra were acquired with the laser for 1 s, and measurements were averaged 30 times.

Instrumentation. Raman measurements were conducted by using a lab-made Raman system. Dark-field/bright-field modules (DF/BF objective lens, light source, and optical condenser) were installed in the microscope (BX40, Olympus). Raman spectral calibration was conducted with both silicon (at 520 cm^{-1}) and standard samples (vibrational fingerprints of acetaminophen). An irradiating laser line (532 nm diode, CNI laser) was delivered to the sample surface through a DF objective lens (MPLAN, 100×, 0.9 NA, Olympus). The spectrometer system used a high-throughput volume phase holographic grating (600 gr/mm) combined with a commercial air-cooled CCD (iDus, 416, Andor).

Database Construction. For each of the 14 substances, 2,000 spectra were acquired from a single position per substance with an exposure time of 0.02 s. Among these, the first 1,000 spectra were used to create the database used for training, validation, and testing. The remaining 1,000 spectra were used to form a secondary test database to evaluate the model's generalization capability.

The obtained spectra were preprocessed through normalization, interpolation, and baseline correction. The normalization was performed using the min-max method, which converts values to a range between 0 and 1. Interpolation was conducted using the interp1d function from scikit-learn.[35] Baseline correction was performed using the Asymmetric Least Squares (ASL) Method.[36] With ASL, the *lambda* was set as 100,000, and p was set as 0.001.

The interpolation was conducted within a Raman shift range of 200−3200 cm^{-1}, uniformly divided at 1 cm^{-1} intervals. Interpolation, particularly, was employed to address pixel variation issues in CCD cameras, which are caused by instrument calibration processes and various factors. This ensures consistency between the CNN model and the Raman spectra.

The preprocessed spectra were averaged using Supplementary Algorithm 1. Each spectrum composing the database (here, each set of 1,000 spectra) is used as X. To create a total of 7,069 spectra, M was set to 1,000. When $N = 1$, the unaveraged spectra are saved. If $N = 2$, the average of the two spectra is calculated and saved. The spectra used for averaging here are randomly selected. N is increased by 1, resulting in repetition up to 1,000. Here, when $N > 1$, a spectrum used once is not redundantly used again for averaging.

The averaged data set was divided into three subsets corresponding to high, medium, and low SNR levels for cross-validation. The 7,069 spectra for each material were divided into three groups of 2,356 spectra each. The low-SNR data set consisted of the 2,357 spectra with the lowest SNR values, the medium-SNR data set contained the next 2,356 spectra, and the high-SNR data set included the remaining 2,356 spectra with the highest SNR values.

Deep Learning Model Training. The deep learning model was developed and trained using PyTorch.[37] For training, the data was divided into 60%, 20%, and 20% for the training, validation, and test sets, respectively. The maximum number of epochs was set to 15, and the model with the highest validation accuracy was selected. The loss function employed was binary cross-entropy loss, and the optimizer used was AdamW.[38] The weight decay was set to 0.006, the learning rate was set to 7e-05, and the batch size was set to 256.

Computation. The interpolation of spectra was conducted by Scikit-learn.[39] For training the CNN models, a Geforce RTX 4070 Ti (NVIDIA, United States) was used for computation. For the conduction of PCC, SCC, ED, MD, and CS, MATLAB R2024a software (MathWorks, United States) was utilized.

RESULTS AND DISCUSSIONS

Raman Spectral Similarity of $-CH_2-$ Chain Molecules. Fatty acids such as SA and OA, which are similar to PE, have a $-CH_2-$ chain structure and exhibit similar Raman spectra. Additionally, SDS shows a similar spectrum due to the molecule structure. Furthermore, oils coming from the environment contain various fatty acids and often exist as mixtures.[40] Therefore, when analyzing samples collected from the environment, it may be necessary to classify them based on data from mixtures rather than individual fatty acids. To address this, samples simulating environmental oils were

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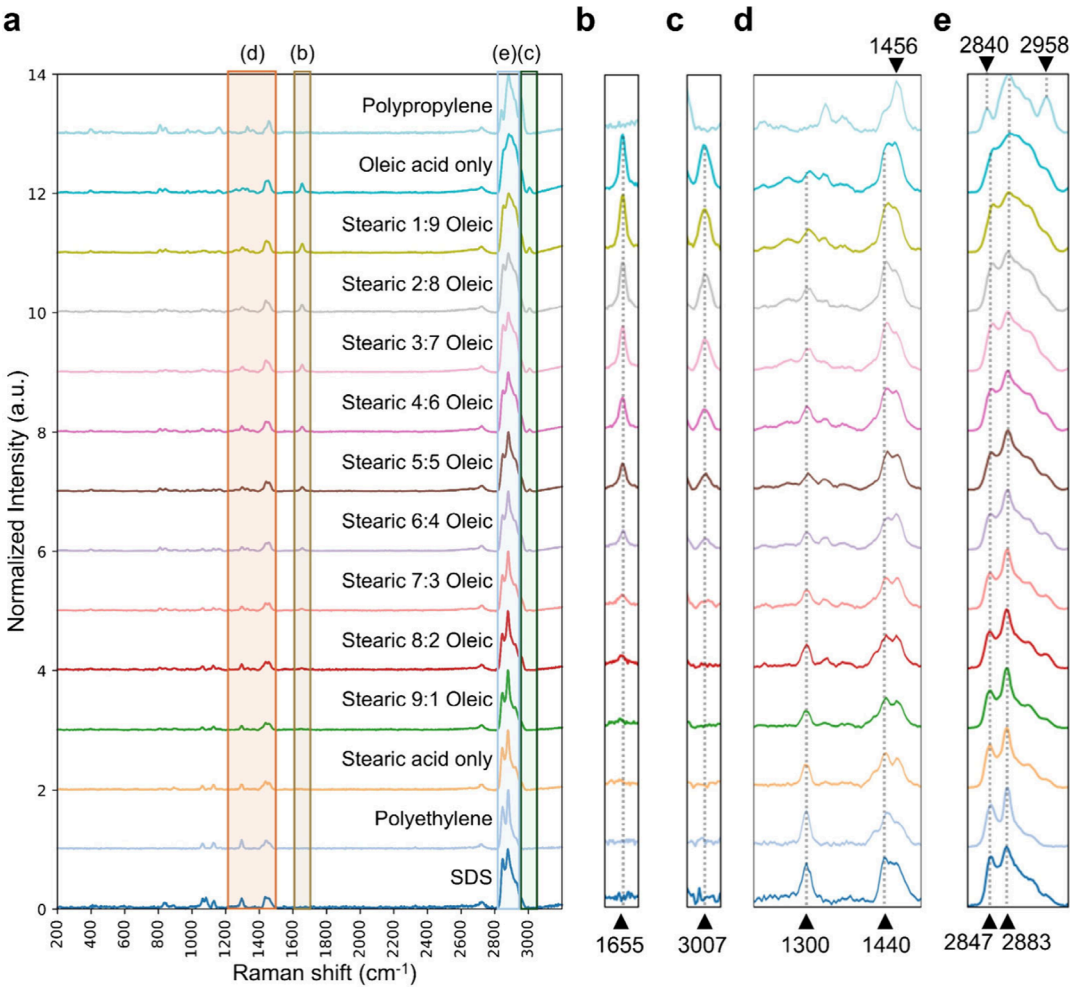


Figure 1. a. Raman spectra of the substances used in the experiment. **b.** Peak intensity changes at 1655 cm^{-1}. **c.** Peak intensity changes at 3007 cm^{-1}. **d.** Peak similarity in the range of 1200−1500 cm^{-1}. **e.** Peak similarity in the range of 2800−3000 cm^{-1}.

prepared by mixing SA and OA in ratios ranging from 9:1 to 1:9.

As shown in Figure 1a, these molecules and mixtures, excluding PP, exhibit highly similar spectra in the Raman shift ranges of 1250−1450 cm^{-1} (Figure 1d) and 2845−2880 cm^{-1} (Figure 1e), which are attributed to the vibrations of the $-CH_2-$ chain. Figure 1b and 1c show the intensity of each material at Raman shifts of 1655 cm^{-1} and 3007 cm^{-1}, respectively. As the proportion of OA increases, the signal intensities at 1655 and 3007 cm^{-1} also increase. These signals are attributed to the C═C bond in OA, with the 1655 cm^{-1} signal corresponding to the vibration of C═C, and the 3007 cm^{-1} signal corresponding to the vibration of H−C═C.[41−43] The additional uniformity verification of the mixture is in Supplementary Note 1.

The resemblances among these spectra may diminish the reliability and accuracy of HQI-based techniques, thereby hindering the classification of microplastic PP and SA.

Reliability Limits for Low SNR Spectra in HQI-Based Classification. As shown in Figure 2a, the Raman spectra of SA and PE exhibit highly similar patterns. This spectral similarity poses a challenge for analyzing environmental samples, where fatty acids can be widely present and may remain in small amounts after pretreatment.

Furthermore, microplastics themselves yield weak Raman signals due to their extremely small size,[44] resulting in a low SNR during analysis. Consequently, while achieving a high SNR requires long exposure times, this significantly increases the overall analysis time when the number of microplastics is large, making large-scale analysis challenging.[26] Therefore, for the quantitative analysis of microplastics, it is essential to conduct large-scale analysis with low exposure times. Under conditions where spectra with low SNR are obtained, methods capable of reliably classifying microplastics are indispensable.

However, correlation coefficient (HQI) based methods can yield ambiguous results[16] for similar Raman spectra, such as SA and PE. Figure 2b and 2c present the results of applying the PCC and SCC methods to SA and PE spectra with various SNRs shown in Figure 2a. Notably, as the SNR decreases, the HQI values tend to decline. This trend can lead to ambiguous results in HQI-based methods, as these methods consider HQI a reliability score and treat values above a specific threshold as trustworthy. If the threshold is set too high, spectra with low SNR can result in unreliable analysis outcomes. Particularly, in the case of SCC, HQI values fall below 0.6 as the SNR decreases, leading to results that are difficult to trust.

The severe decrease in HQI for SCC with low-SNR spectra may stem from the rank-based method used for calculation (Supplementary Note 6). Under low SNR conditions, strong noise can distort the intensity ranking of spectral peaks, causing a mismatch with the rank of the reference spectrum. While PCC evaluates the direct difference between intensity

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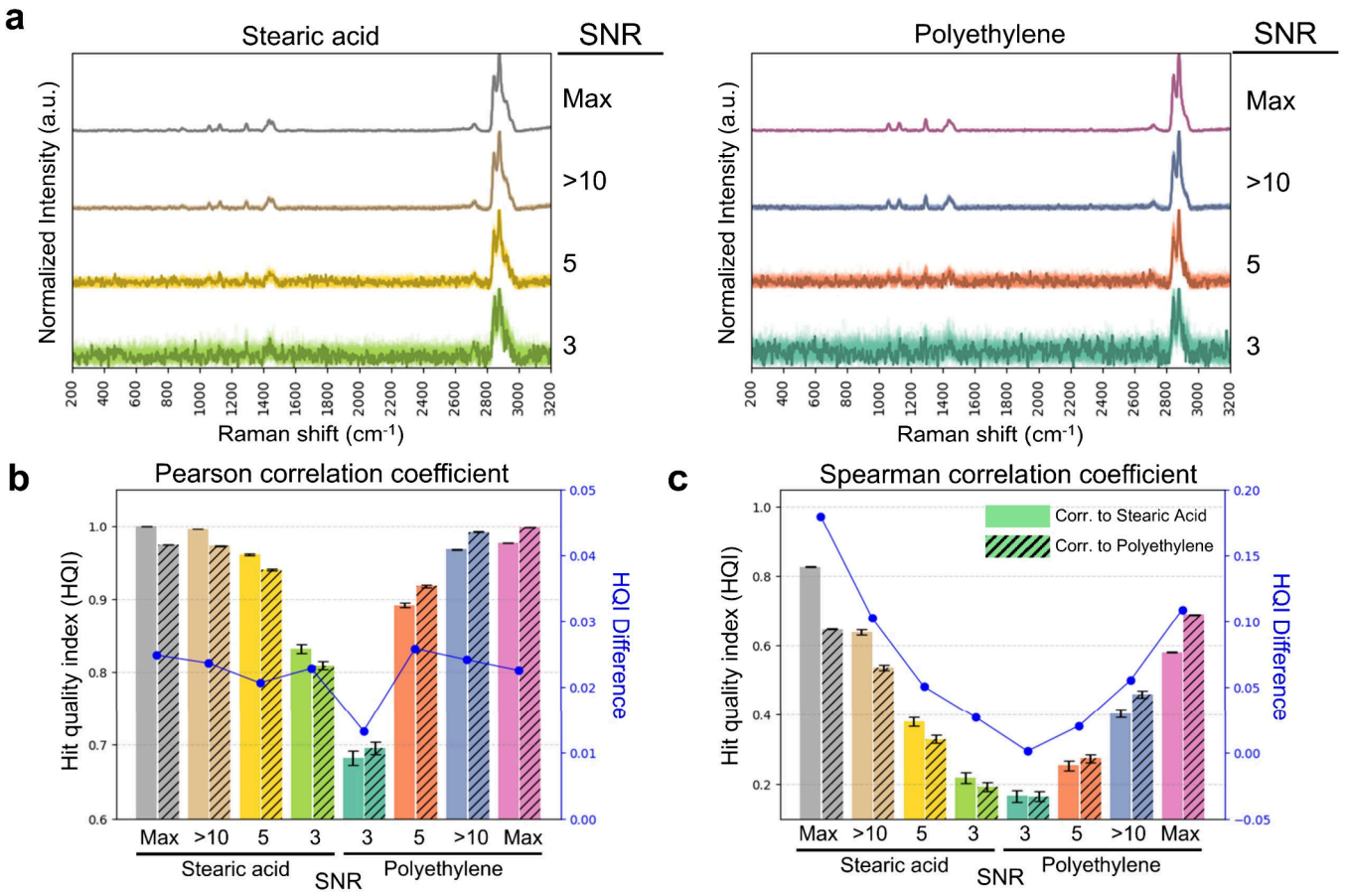


Figure 2. a. Raman spectra of Stearic acid (left) and Polyethylene (right) in the data set that was used for training the DL model. Each spectrum shows its SNR value. All spectra were normalized. The calculation of SNR is explained in Supplementary Note 2. **b, c.** Results of the Pearson/Spearman correlation coefficient for the standard spectra of Stearic acid and Polyethylene for each of the spectra in **a** (bar graph) and the difference between the respective values (line graph). The bars without dashes represent the correlation coefficients calculated for standard stearic acid, while the bars with dashes represent those calculated for standard polyethylene. Corr. indicates correlation. The error bars represent the 95% confidence interval.

values at each Raman shift, it can still exhibit high accuracy even for spectra with low SNR. These differences critically affect HQI computation, ultimately leading to lower HQI values for SCC.

In the case of PCC, the HQI differences between standard SA and PE spectra range from 0.04 to 0.02 at most, which corresponds to only a 4% to 2% difference when converted to percentages. This indicates that the HQIs for standard SA and PE are not significantly different, making the results potentially unreliable. Moreover, high HQI values can introduce substantial flaws in result selection.

Consequently, due to the similar spectral features of SA and PE, the use of correlation coefficient methods can lead to reliability issues, more severely for spectra with low SNR, which are frequently observed in microplastics. This problem makes it challenging to classify PE, fatty acids, and SDS, posing difficulties in the classification of microplastics.

Thus, we employed a DL model that is more reliable and accurate than the HQI-based technique. To develop a deep learning model for qualitative and quantitative analysis of the five substances and the SA/OA mixtures, the output was binarized as shown in Figure 3.

Binary Encoding for Mixture Classification. When performing deep learning-based classification using Raman spectra, one-hot encoding is typically employed.[23,45,46] In this

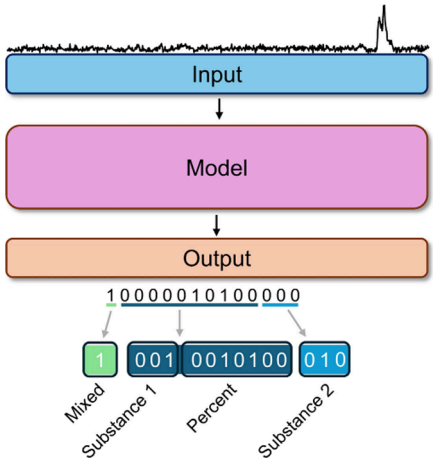


Figure 3. Scheme for the input and output of the model. The input of the model is a Raman spectrum, and the output is a binarized 14-digit binary number composed of 0s and 1s. The first 11 digits represent the first substance, while the last 3 digits represent the second substance. The first 11 digits include a single bit indicating the presence of a mixture, three bits representing the substance type, and the final seven bits representing the percentage of mixing. The last 3 digits indicate only the substance type. Detailed explanations for each binary digit are provided in Table 1.

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encoding, each part of the output tensor corresponds to a specific substance, and if the model classifies a given value as 1, it indicates that the substance is present; if classified as 0, it indicates the substance is absent. However, as the number of substances to classify increases, the size of the output tensor grows proportionally, leading to scalability issues.

To address this problem, the binary encoding method is employed, ensuring scalability for the classification of a broader range of substances. By applying binary encoding, the number of substances that can be represented increases significantly. For instance, with a 2-bit binary number, four substances can be represented as 00, 01, 10, and 11. In contrast, a 10-bit binary number can represent 1,024 substances, with the number of representable substances increasing at an exponential rate (2^n) with respect to the number of bits, n. Thus, binary encoding enables the representation of more substances within a limited tensor size. Especially, the further experiments comparing binary encoding and one-hot encoding with RRUFF[47] showed that the binary encoding can make a more compact model and potentially increase accuracy (Supplementary Note 3).

In this experiment, as shown in Figure 3, the input data to the model is the Raman spectrum, and the output is represented as binary values. The model generates a 14-bit binary output, with the first 11 bits representing the first substance (Substance 1) and the last 3 bits representing the second substance (Substance 2).

Within the bits representing the first substance, the first bit indicates whether the substance is a mixture. If the binary value is 0, the substance is not a mixture; if it is 1, it indicates a mixture. The second to fourth bits represent the material type, allowing for the representation of up to 8 substances. For the second substance, the 4 bits for mixture and material type are used. In this experiment, five substances were assigned unique binary codes, with "None" designated as 101, 110, and 111. The binary digits following the fifth bit indicate the mixing ratio of the substance. For pure substances, this value corresponds to 100% and is represented as '110 0100'. For mixtures, the value varies according to the mixing ratio. Detailed interpretations of these binary digits are summarized in Table 1.

To train the model, the labels of each spectrum were encoded as binary values. These binary values were set as the target output of the model, and training was performed using a loss function. Once trained, the model generates binary outputs, which are decoded to evaluate the performance of the model.

Table 1. Translation of Binarized Output

Type	Binary code	Translation
Mixed	0	Negative
	1	Positive
Substance	000	Polyethylene
	001	Stearic acid
	010	Oleic acid
	011	Sodium dodecyl sulfate
	100	Polypropylene
	101, 110, 111	None
Mixed Percent	000 0000	0%
	000 1010	10%
	001 0100	20%
	001 1110	30%
	⋮	⋮
	110 0100	100%

To train and evaluate the prepared model, a large amount of diverse spectra for each substance needs to be prepared. To achieve this, numerous spectra with SNR 3 were collected using short exposure times, and then augmentation of the spectral data was performed through an averaging method.

Averaging Raman Spectra for Database and U-net Model. For the prepared substances shown in Figure 1, Raman spectra acquired 0.02-s exposure time were averaged as shown in Figure 4a. The averaging method was proceeded by following the given formula described in the Experimental Section. As n increases, the SNR of the spectra progressively improves, demonstrating a pattern where the spectral signal becomes distinct. Consequently, this averaging method enables the construction of a Raman spectrum database with various SNR levels. The spectra with the lowest and highest SNR in the averaged database are in Figure S8−S9.

The averaged spectra were organized into a database for each material (Figure 4b), forming a total of 98,966 spectra across 14 materials collectively referred to as the "Averaged Database". This database was subsequently used for training the model with the structure shown in Figure 4c.

U-Net is designed with a U-shaped architecture that processes input data through the down-sampling and up-sampling stages. In the down-sampling stage, convolution and pooling extract key features, while the up-sampling stage uses deconvolution and skip connections to preserve critical elements from earlier stages. These characteristics enable U-Net to retain initial spectral judgments while incorporating the insights derived from the sampling stages to form the output.

Consequently, input data was processed using the U-Net architecture and then flattened into a one-dimensional format. Fully connected layers processed the data further to produce the binarization output. The model structure comprises a U-Net architecture combined with fully connected layers. Detailed model architecture and training results are provided in Figures S10−14.

To compare the effectiveness of our U-Net model, accuracy comparisons were made with a simple CNN, ResNet,[48] and U-Net with multihead attention and positional encoding[49] (Supplementary Note 4). As a result, it was confirmed that the compared models had lower accuracy than not only our U-Net model but also PCC. Additionally, using Grad-CAM[50] and SHAP,[51] the regions on which the trained U-Net model was focusing were identified (Supplementary Note 5). Overall, it was confirmed that the model predominantly utilized key Raman shifts (2844 cm^{-1} and 2884 cm^{-1}) for its analysis, indicating that it learned patterns consistent with domain knowledge.

Performance Evaluation of Deep Learning Versus HQI-Based Techniques. To compare the ability for classification of Raman spectra with the trained U-net deep learning (DL) model, we utilized the 5 HQI-based techniques, illustrated in Figure 5a−e: PCC, SCC, CS, ED, and MD. The calculation processes and definitions of these techniques are provided in Supplementary Note 6.

Figure 5f−g demonstrates the accuracy comparison between the DL model trained on our methodology and the similarity comparison methods for the secondary test database. The secondary test database was acquired using with same process as the database for the DL training and constructed using the

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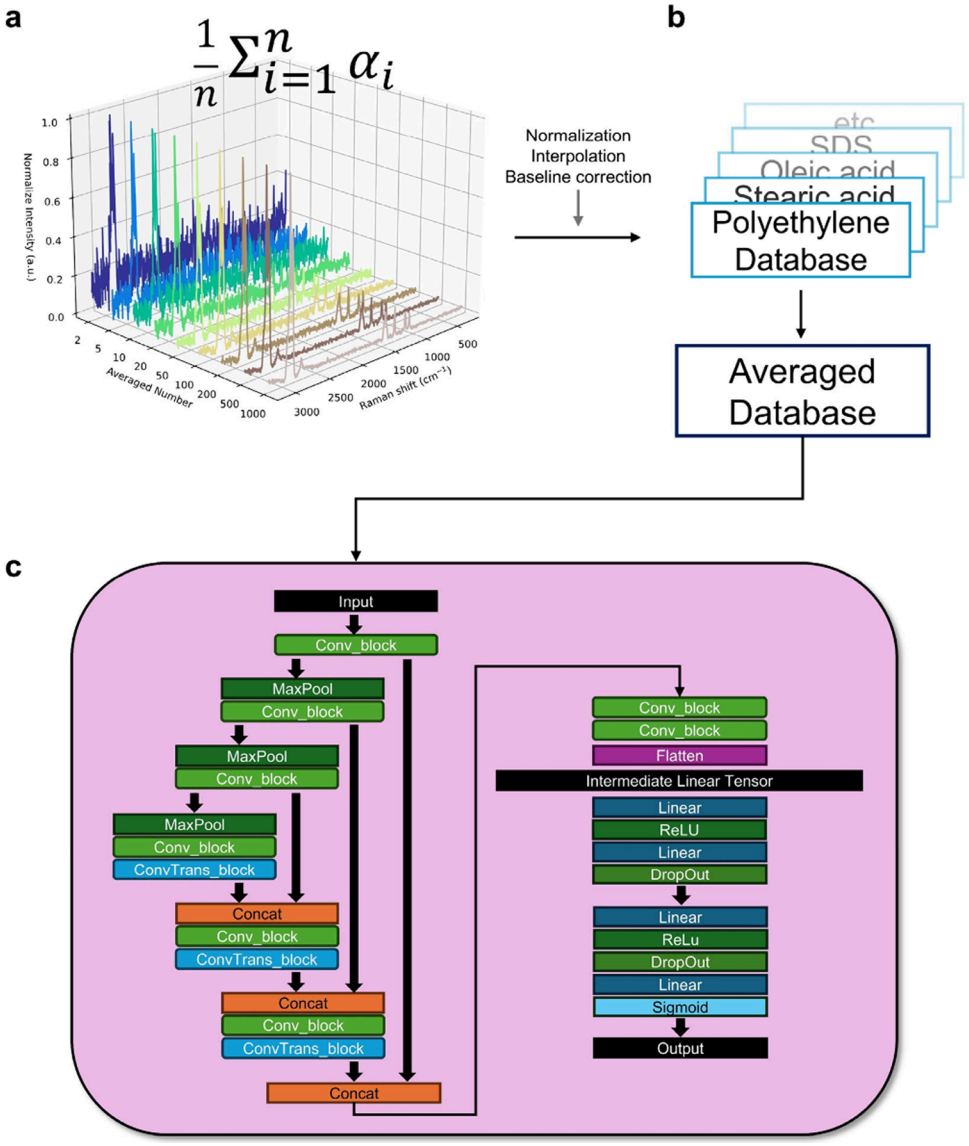


Figure 4. a. Results of averaging polyethylene spectra. α_i represents each acquired Raman spectrum, and n denotes the average number, indicating how many spectra are averaged. As n increases, the SNR of the spectra improves. The averaged spectra are preprocessed with normalization, interpolation, and baseline correction. **b.** The averaged spectrum database for each material constitutes the "Averaged Database". **c.** Simplified structure of the CNN model. The input with positional encoding passes through a U-shaped structure, is then flattened into a one-dimensional form, and subsequently processed through fully connected layers to produce a 22-bit binary output.

same preprocessing and averaging methods as the training database, as detailed in the Experimental Section. This secondary test database was prepared separately to determine the generalization degree of each method. The confusion matrix is shown in Figure S15.

For HQI comparisons, we used spectra with the highest SNR values, averaged with a count of 1,000, as representative spectra for each material in the database used for the DL model. A total of 14 spectra were used as the database for similarity comparisons.

Generally, various HQI thresholds are used in similarity comparisons,[15] ranging from 0.6 to 0.85. In the case of PCC, SCC, and CS, we set HQI thresholds in increments of 0.1 (10%) and performed HQI calculations accordingly. If the HQI of the most similar material exceeded the set threshold, the material was confirmed as classified and used to calculate accuracy. The accuracy of ED and MD was calculated by selecting the substance that showed the minimum HQI, without a threshold.

As depicted in Figure 5f, the DL model achieved a 2.05% accuracy improvement over PCC with an HQI of 0.6 (PCC60) and an 11.09% improvement over PCC with an HQI of 0.9 (PCC90) for all SNR spectra. Furthermore, it showed improvements of 61.44% over SCC at an HQI of 0.6 (SCC60), up to 50.51% over CS, 36.69% over ED, and 51.12% over MD. The overall accuracy is in Table S6.

As shown in Figure 5g, the nonaveraged spectra are additionally examined to check the ability for classification of spectra with high noise. The DL model demonstrated a 21.21% accuracy improvement over PCC with an HQI of 0.6 (PCC60) and a 48.97% improvement over PCC with an HQI of 0.9 (PCC90). Other methods besides PCC did not exceed 20% accuracy.

This trend can also be checked from the analysis results of the secondary test database, which was divided into three

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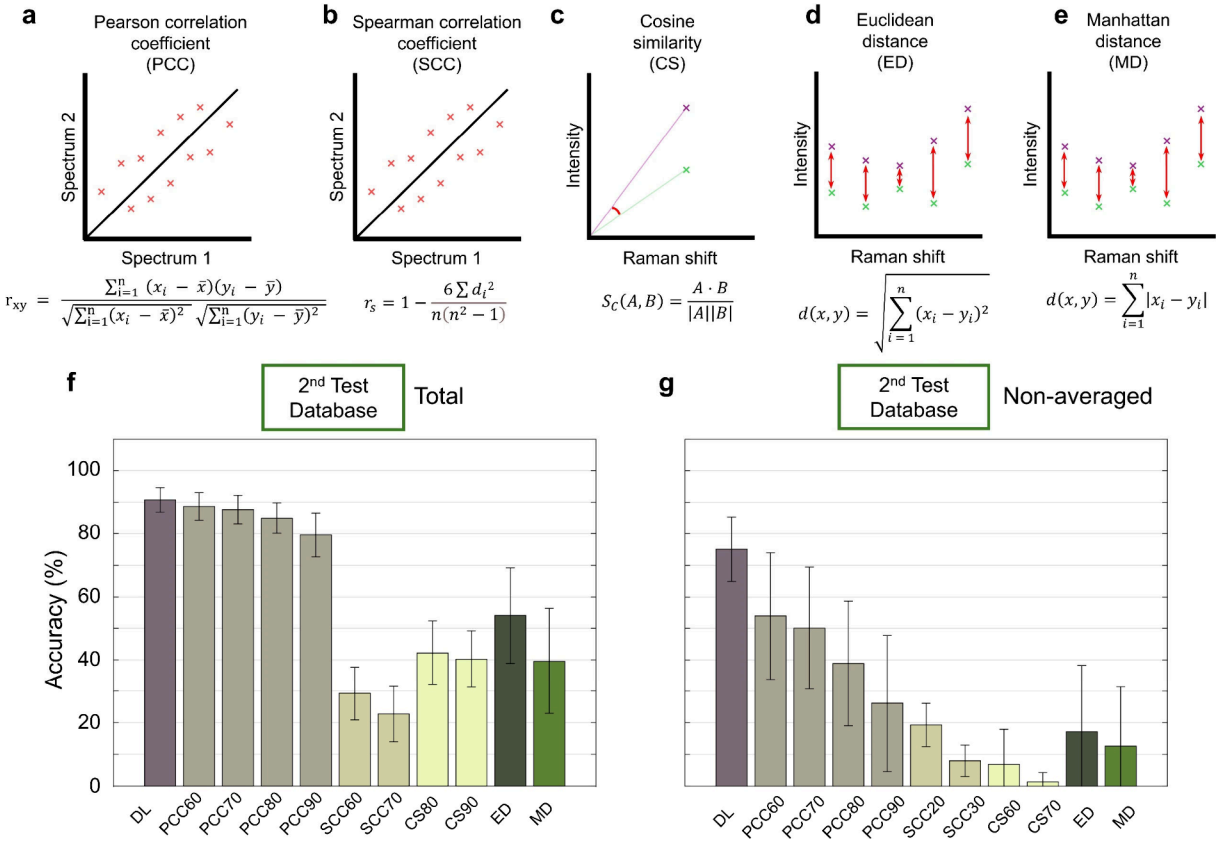


Figure 5. a–e. Schematic representations and formulas of various HQI-based techniques (Pearson correlation coefficient (PCC), Spearman correlation coefficient (SCC), cosine similarity (CS), Euclidean distance (ED), and Manhattan distance (MD)). **f–g.** Comparison of accuracy between the deep learning (DL) model and HQI-based methods. **f.** Accuracy comparison of each method based on the entire spectrum of the secondary test database. **g.** Accuracy comparison for nonaveraged spectra in the secondary test database. The error bars represent the 95% confidence interval.

groups (high, medium, and low) based on SNR. In Figure S16, while nearly all methods exhibit high accuracy for high SNR spectra, the accuracy of the methods declines as the SNR decreases, with DL showing the smallest decrease among them.

This is because HQI calculation methods struggle to accurately classify due to the presence of noise in the signal. In contrast, the DL model automatically extracts and learns features from Raman spectra of various SNR levels, enabling accurate classification. This confirms that the DL model outperforms HQI comparison methods not only for high-SNR spectra but also for spectra with high noise. As a result, the DL model is suitable for the classification of Raman spectra, which have potential bias,[52] such as noise effect, detector effects, instrumental effects, etc. The overall accuracy is in Table S7.

To explain the strength of the DL model for nonaveraged spectra, the model's accuracy was evaluated the cross-validation based on varying SNR levels (Supplementary Note 7). Here, a model trained on low SNR spectra achieves approximately 78% accuracy for spectra covering a broad range of SNR values. However, as the SNR of the training spectra increases, the model shows a tendency to fail in accurately classifying low SNR spectra. Consequently, it is evident that training the model exclusively with spectra of a specific SNR results in lower accuracy and a poorer degree of generalization compared to training with a spectral database encompassing a diverse range of SNRs.

Beyond its accuracy, the DL model shows advantages in computational efficiency. Supplementary Note 8 provides a detailed comparison, revealing that the primary difference lies in their operational structure. The HQI-based technique performs calculations for each spectrum against a reference database, a process whose time cost grows with both the database size and the number of spectra. Conversely, the DL model undergoes a one-time training phase where it learns the features of the reference data.

Our analysis shows that post-training model allows for extremely rapid classification, resulting in approximately 21.3 times faster than the HQI-based technique. Consequentially, once the analytical task involves approximately 35,000 spectra or more, the cumulative time for the DL approach becomes less than that for the HQI method, highlighting its suitability for the large data sets typical of microplastic analysis.

Moreover, to shorten the test process with the secondary test set, bootstrapping can be conducted (Supplementary note 9).

Classification of Microplastics and Fatty Acids. Here, we verified the model's fitness for microplastics and fatty acids. Figure 6 shows the spectra of particles simulating environmental samples in which PE microplastics are dispersed together with olive oil. Table 2 presents the results of each analytical method applied to the spectra in Figure 6. The preparation method for this sample is described in the Experimental Section.

The spectra of particles #1–6 exhibit spectra similar to OA. Similar spectral features are also observed for particles #15–24, as olive oil contains approximately 70–80% OA.[33,34] In

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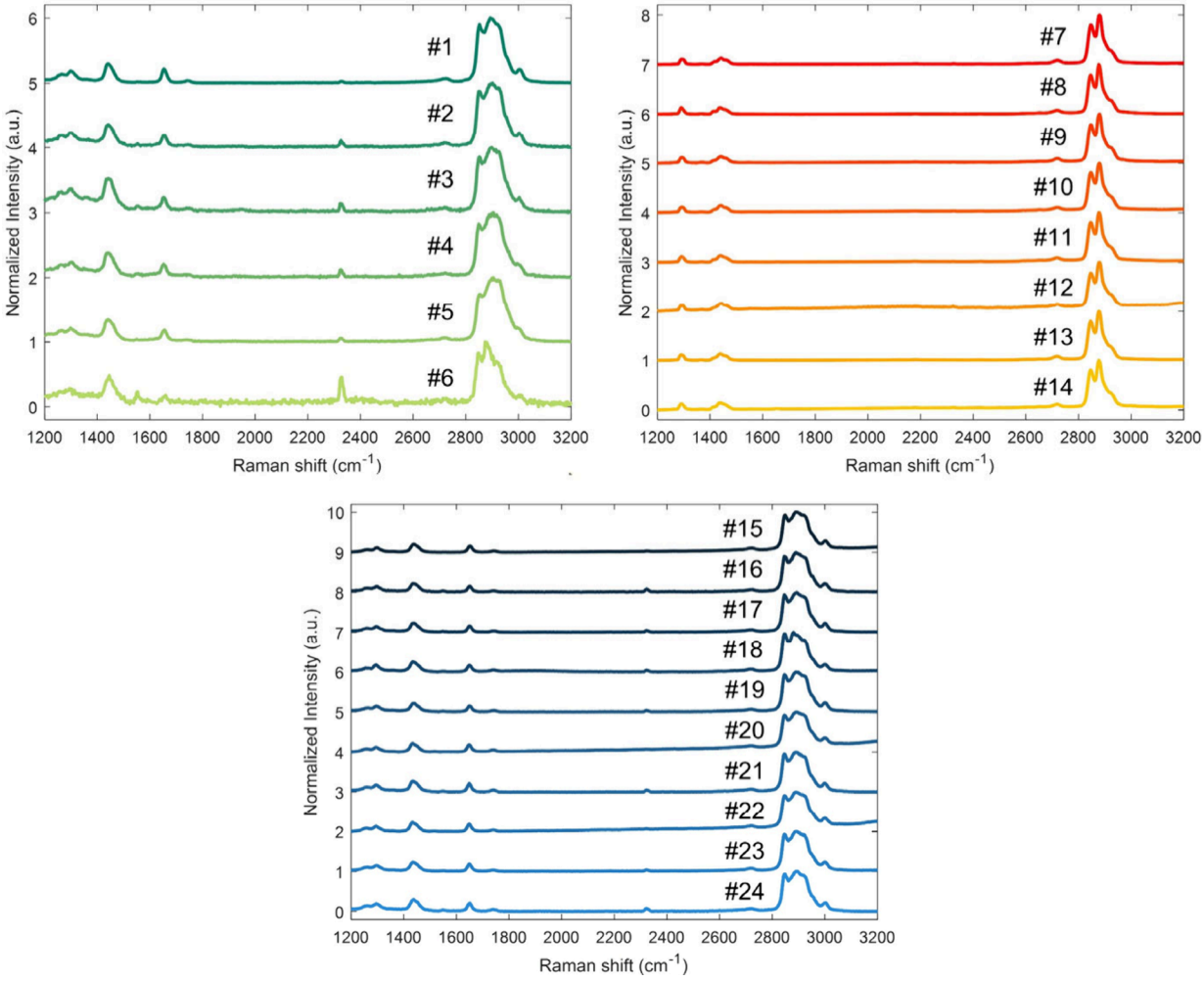


Figure 6. Raman spectra of PE microplastics (#7−#14) and olive oil particles (#1−#6, #15−#24). The 200−1200 cm^{-1} region was removed due to strong signals from the silicon wafer. The Bright-field microscopy images of particles are shown in Figure S17, and spectra including the silicon wafer signals are shown in Figure S18.

contrast, particles #7−14 show spectra resembling PE, which is consistent with the fact that PE particles were imaged. Additionally, the peak of 2327 cm^{-1} is derived from N_2 in the air.[53]

The results in Table 2 show that the analysis by DL is highly consistent and more accurate compared to other HQI-based techniques. Specifically, the DL analysis result of 3:7 for olive oil indicates that the sample consists of 30% SA and 70% OA, which closely aligns with the known OA content of 70−80% in olive oil. In contrast, the HQI techniques lacks consistency—in PCC method, most PE particles (#7−14) were misclassified as SA despite being confirmed as PE particles, and even the olive oil sample (#1−6) was misclassified as pure OA.

Additionally, as shown in Figure S18, the influence of the silicon wafer was observed during spectrum acquisition, and this substrate signal significantly reduced the accuracy of the HQI techniques, as demonstrated in Table S8. In contrast, the DL method, despite including signals from the silicon wafer, successfully performed correct classification, demonstrating its applicability to environmentally simulated samples.

CONCLUSIONS

Consequently, our U-net based deep learning model can accurately classify and analyze Raman spectra from individual substances and their complex mixtures. The current protocols for microplastic analysis, which involve sample pretreatment-spectrum acquisition-spectrum analysis, stand to gain significantly from the integration of such a model. By incorporating our DL model into the spectrum analysis stage, laboratories can achieve more standardized and reproducible results. The architecture of model allows for easy deployment, reuse, and fine-tuning, ensuring its flexible adaptation to specific laboratory data and workflows.

A primary challenge in microplastic analysis is the accurate identification of components in spectra with a low SNR, a scenario where traditional methods like HQI comparison often weaken. We have shown that our U-net model provides superior accuracy under these conditions.

Furthermore, by output binarization, our model encapsulates more detailed information, proving effective for the classification of both microplastics and fatty acids. While the model shows great potential, its application to highly complex mixtures presents a challenge due to the exponential increase in potential substance combinations. To address this, future work could focus on constructing a comprehensive spectral database of diverse mixtures. Alternatively, data augmentation through artificial synthesis of spectra could be employed to improve the generalization capability of the model.

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Table 2. Classified Results of PE Particles and Oil from DL and HQI Techniques[a]

		DL	PCC		SCC		CS		ED	MD
#	GT	Result	Result	HQI	Result	HQI	Result	HQI	Result	Result
1	3:7	3:7	OA	0.97	1:9	0.63	OA	0.98	OA	OA
2	3:7	3:7	OA	0.96	1:9	0.51	OA	0.96	OA	SA
3	3:7	3:7	OA	0.91	1:9	0.40	OA	0.92	OA	OA
4	3:7	3:7	OA	0.95	1:9	0.50	OA	0.96	OA	OA
5	3:7	3:7	OA	0.95	1:9	0.56	OA	0.96	OA	OA
6	3:7	3:7	4:6	0.92	1:9	0.36	SDS	0.92	SDS	SDS
7	PE	PE	PE	0.98	SA	0.59	PE	0.99	SDS	SA
8	PE	PE	PE	0.98	SA	0.62	PE	0.98	SDS	1:9
9	PE	PE	SA	0.98	SA	0.58	SA	0.99	SDS	SA
10	PE	PE	SA	0.98	SA	0.57	SA	0.98	SA	SDS
11	PE	PE	SA	0.98	SA	0.58	SA	0.98	SA	SA
12	PE	PE	SA	0.96	SA	0.50	SDS	0.93	SDS	SDS
13	PE	PE	PE	0.98	SA	0.60	PE	0.98	SA	SA
14	PE	PE	SA	0.98	SA	0.56	SA	0.98	SDS	SDS
15	3:7	3:7	1:9	0.97	1:9	0.69	1:9	0.97	OA	OA
16	3:7	3:7	1:9	0.98	1:9	0.60	1:9	0.8	1:9	1:9
17	3:7	3:7	1:9	0.97	1:9	0.63	1:9	0.98	1:9	1:9
18	3:7	3:7	1:9	0.98	OA	0.60	2:8	0.98	1:9	1:9
19	3:7	3:7	1:9	0.98	1:9	0.66	1:9	0.98	1:9	1:9
20	3:7	3:7	1:9	0.94	7:3	0.60	OA	0.94	OA	OA
21	3:7	3:7	1:9	0.97	1:9	0.63	1:9	0.98	1:9	1:9
22	3:7	3:7	1:9	0.95	2:8	0.63	OA	0.95	OA	OA
23	3:7	3:7	1:9	0.98	1:9	0.70	1:9	0.98	1:9	1:9
24	3:7	3:7	1:9	0.97	1:9	0.55	1:9	0.97	1:9	1:9

[a]n:m indicates $n \times 10\%$ SA and $m \times 10\%$ OA. In the case of HQI techniques, 1200−3200 cm^{-1} of Raman shift is used. GT indicates ground truth.

In conclusion, the U-net deep learning approach presented here offers a robust and accurate method for spectral analysis. By overcoming the limitations of traditional techniques with low SNR data, our model contributes to a faster, more reliable, and ultimately more standardized methodology for microplastic analysis.

■ ASSOCIATED CONTENT

Data Availability Statement

The data sets generated and analyzed in the present study are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.5c00584.

Supplementary Algorithm 1. Pseudo code for spectrum augmentation based on averaging. **Supplementary Note 1.** The Uniformity of the SA and OA Mixture. **Figure S1.** Verification of the homogeneity of the SA:OA mixtures. **Supplementary Note 2.** Calculation of Signal-to-Noise Ratio. **Figure S2.** SNR calculation process scheme for polyethylene Raman spectrum. **Supplementary Note 3.** Ability of the Output Binarization. **Figure S3.** Binary encoded and one-hot encoded model architectures. **Supplementary Note 4.** Classification Ability of Other DL Model Structures. **Figure S4.** A simple CNN model and ResNet model structure. **Supplementary Note 5.** Model Interpretation through Grad-CAM and SHAP. **Figure S5.** Grad-CAM result for the U-net Model. **Figure S6.** SHAP result for the U-net Model. **Supplementary Note 6.** Calculation of HQI measurements. **eq 1.** The formula of the Pearson correlation coefficient. **eq 2.** The formula of the Spearman correlation coefficient. **eq 3.** The formula of the Euclidean distance. **eq 4.** The formula of the Manhattan distance. **eq 5.** The formula of the cosine similarity. **Supplementary Note 7.** Cross Validation of High/Medium/Low SNR Data set. **Table S1.** Cross-validation results on the secondary test data set. **Supplementary Note 8.** Time efficiency of DL and HQI-based techniques. **Table S2.** Training time for each num_worker value in the DL model. **Table S3.** Computation time for the secondary test data set. **Figure S7.** The total computation time varies depending on the number of spectra. **Supplementary Note 9.** Bootstrapping for Secondary Test Data set. **Table S4.** The accuracy results for each iteration of the secondary test data set. **Figure S8.** The lowest SNR spectra in the acquired database of Figure 3. **Figure S9.** The highest SNR spectra in the averaged database of Figure 3. **Figure S10.** The structure of the model. **Figure S11.** The accuracy and loss graph for every epoch of the model. **Figure S12.** The confusion matrix for the train set of the selected epoch. **Figure S13.** The confusion matrix for the validation set of the selected epoch. **Figure S14.** The confusion matrix for the test set of the selected epoch. **Figure S15.** The confusion matrix for the secondary test set of the selected epoch. **Figure S16.** Additional comparison of accuracy between the deep learning (DL) model and HQI-based methods. **Figure S17.** Bright-field microscopy images of PE microplastics and olive oil particles. **Figure S18.** Raman spectra in the 200−3200 cm^{-1} range for PE microplastics and olive oil particles. **Table S5.** Peak assignment of the selected substances Raman spectrum. **Table S6.** The accuracy

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results for the total spectra of the secondary test database. **Table S7.** The accuracy results for non-averaged spectra of the secondary test database. **Table S8.** Classified result of PE particles and oil from DL and HQI methods with 200−3200 cm^{-1} of Raman shift. **Table S9.** Model tensor size. (PDF)

AUTHOR INFORMATION

Corresponding Author

Dongha Shin − *Department of Chemistry and Chemical Engineering, Inha University, Incheon 22212, Republic of Korea; Program in Biomedical Science and Engineering, Inha University, Incheon 22212, Republic of Korea; NanoRaman Analysis Corp., Incheon 22212, Republic of Korea;* orcid.org/0000-0003-3939-6797; Email: dongha@inha.ac.kr

Authors

Jeonghyun Lim − *Department of Chemistry and Chemical Engineering, Inha University, Incheon 22212, Republic of Korea;* orcid.org/0009-0006-5116-5157

Juhui Seo − *Department of Chemistry and Chemical Engineering, Inha University, Incheon 22212, Republic of Korea*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.analchem.5c00584

Author Contributions

Jeonghyun Lim: Formal analysis, Validation, Writing—review and editing. **Juhui Seo:** Environmental sample analysis. **Dongha Shin:** Conceptualization, Supervision, Writing—original draft, Formal analysis, Validation, Writing—review and editing.

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Notes

The authors declare no competing financial interest.

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